Comparison of C and C++ & Introduction to Object Oriented Concepts

C and C++

C	C++
C was developed by Dennis Ritchie between the year 1969 and 1973 at AT&T Bell Labs.	C++ was developed by Bjarne Stroustrup in 1979.
C does no support polymorphism, encapsulation, and inheritance which means that C does not support object oriented programming.	C++ supports polymorphism, encapsulation, and inheritance because it is an object oriented programming language.
C is a subset of C++.	C++ is a superset of C.
C contains 32 keywords.	C++ contains 63 keywords.
For the development of code, C supports procedural programming.	C++ is known as hybrid language because C++ supports both procedural and object oriented programming paradigms.
Data and functions are separated in C because it is a procedural programming language.	Data and functions are encapsulated together in form of an object in C++.

C and C++

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C does not support information hiding.	Data is hidden by the Encapsulation to ensure that data structures and operators are used as intended.
C is a function driven language because C is a procedural programming language.	C++ is an object driven language because it is an object oriented programming.
Function and operator overloading is not supported in C.	Function and operator overloading is supported by C++.
Functions in C are not defined inside structures.	Functions can be used inside a structure in C++.
Header file used by C is stdio.h.	Header file used by C++ is iostream.h.
Virtual and friend functions are not supported by C.	Virtual and friend functions are supported by C++.
C does not support inheritance.	C++ supports inheritance.
C provides malloc() and calloc() functions for dynamic memory allocation, and free() for memory de-allocation.	C++ provides new operator for memory allocation and delete operator for memory de-allocation.

C and C++

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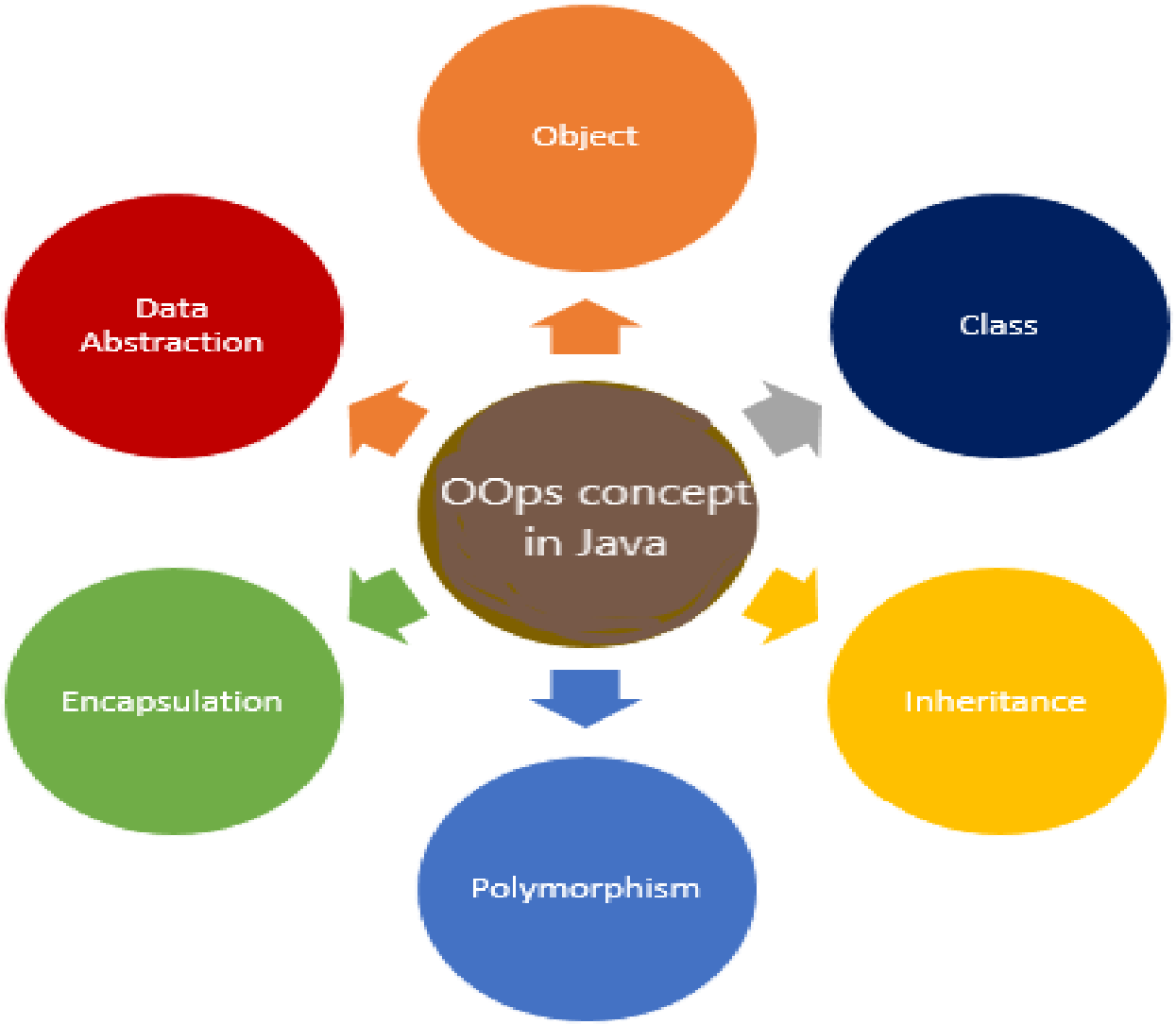
Direct support for exception handling is not supported by C.	Exception handling is supported by C++.
scanf() and printf() functions are used for input/output in C.	cin and cout are used for input/output in C++.
C structures don't have access modifiers.	C ++ structures have access modifiers.
C follows the top-down approach	C++ follows the Bottom-up approach
There is no strict type checking in C programming language.	Strict type checking in done in C++. So many programs that run well in C compiler will result in many warnings and errors under C++ compiler.
C does not support overloading	C++ does support overloading

Features of OOPS



OOCP Concepts

- Class
- Objects
- Data Abstraction
- Encapsulation
- Inheritance
- Polymorphism
- Dynamic Binding
- Message Passing

Steps of Program Creation and Execution

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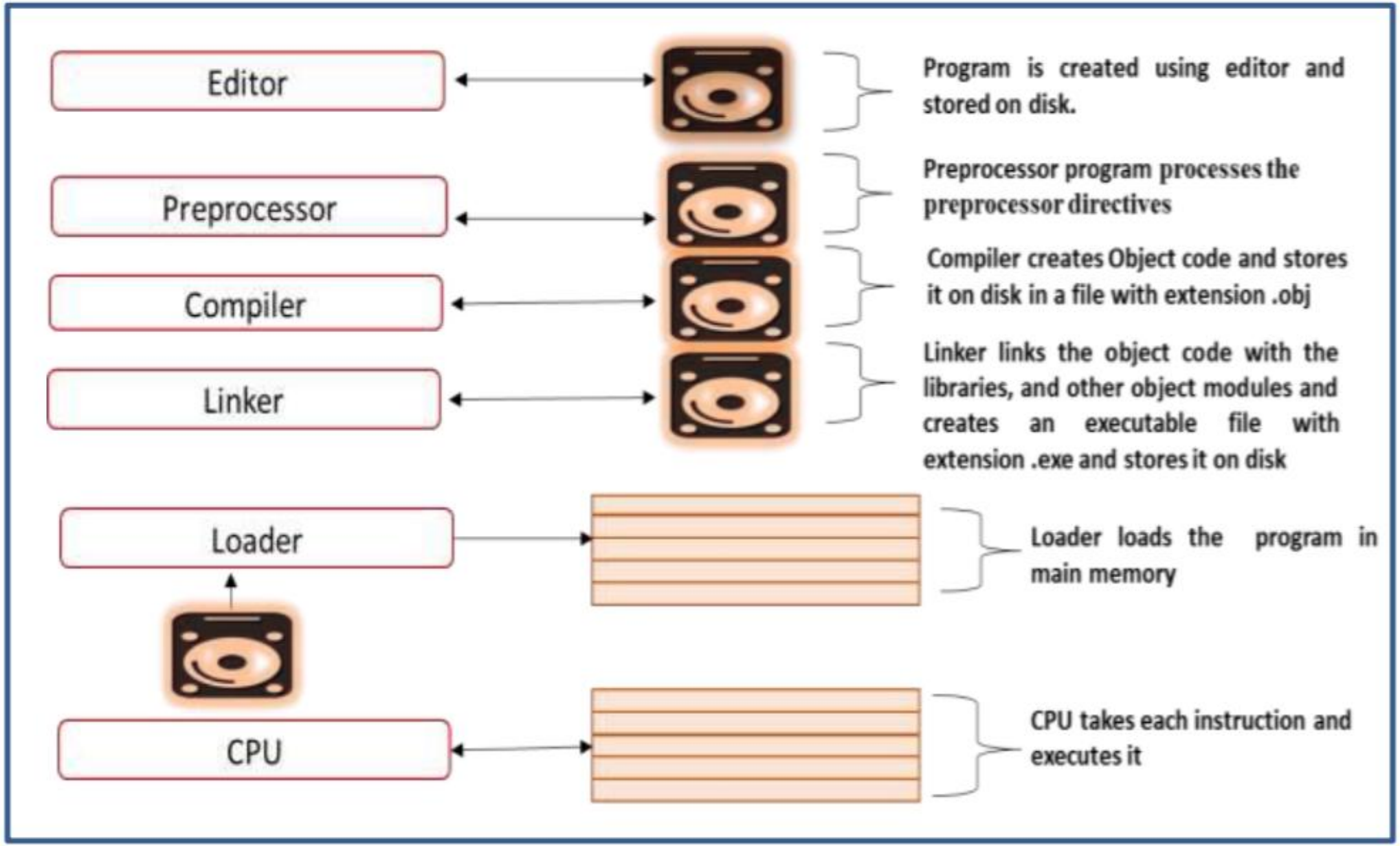

The general structure of C++ program

- 1. Documentation Section
- 2. Preprocessor Directives / Compiler Directives Section Contains file inclusion directives, Macros etc.
- 3. Global Declaration Section
- 4. Class declaration or definition Section
- 5. C++ program entry point - function main ( )
- 6. Beginning of the main function: Left brace { (i) Object/Variables declaration part (ii) Executable statements
- 7. End of the main program: Right brace}
- 8. Other user defined functions if needed

Basic Program & components of Program

```
// Program : Saying Hello with C++
#include<iostream>
using namespace std;
void main()
{
        cout << “Hello World ! I have begun my journey of learning
         Object oriented concepts and programming using C++” ;
}
// note the semicolons; they terminate statements
/* main( ) is a function that takes no arguments ( ) and returns
an int (integer value) to indicate success or failure */
```

Compiling and Executing C++ Program

- 1. Borland C++ / Turbo C++
- 2. Visual C++ [Microsoft Platform]
- 3. Dev C++
- 4. GCC
- 5. Eclipse

Character sets

- The basic source character set consists of 96 characters: the space character, the control characters representing horizontal tab, vertical tab, form feed, and new-line, plus the following 91 graphical characters:14
- a b c d e f g h i j k l m n o p q r s t u v w x y z
- A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
- 0 1 2 3 4 5 6 7 8 9
- _ { } [ ] # ( ) < > % : ; . ? * + - / ^ & | ~ ! = , \ " ’

Escape Sequences

- Nonprintable characters also known as execution characters can be represented by an escape sequence.

Escape Sequence	Character Represented
\a	Alert (bell, alarm)
\b	Backspace
\f	Form feed (new page)
\n	New-line
\r	Carriage return
\t	Horizontal tab
\v	Vertical tab
\’	Single quotation mark
\"	Double quotation mark
\?	Question mark
\\	Backslash

Data Types

- Data type define the way you store the values used by and created by the program.

Group	Type	Size / precision
Character types	char	Exactly one byte in size. At least 8 bits
Integer types	int	Not smaller than short. At least 16 bits
Floating-point types	float double	Exactly 4 byte in size. At least 8 bits
Void type	Void	no storage
boolean	bool	

keywords

asm	**do**	**if**	**return**	typedef
auto	**double**	inline	**short**	*typeid*
bool	*dynamic_ cast*	**Int**	**signed**	*typename*
break	else	**long**	**sizeof**	**union**
case	**enum**	*mutable*	static	**unsigned**
catch	*explicit*	*namespace*	*static_cast*	*using*
char	*export*	new	**struct**	virtual
class	**extern**	operator	**switch**	**void**
const	*false*	private	template	**volatile**
const_cast	**float**	protected	this	*wchar_t*
continue	**for**	public	throw	**while**
default	friend	**register**	*true*	
delete	**goto**	*reinterpret_cast*	try	